Merwyn Kumar Kollabathula

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.30
Bhashyam junior college	Class XII (Board of Intermediate Education Andhra Pradesh)	2024	91.70%
J. Sikile school	Class X (Central Board of Secondary Education)	2022	90.00%

POSITIONS OF RESPONSIBILITY

Welfare Secretary - Students' Gymkhana Council

Elected as the Welfare Secretary to represent the student body and lead campus-wide welfare initiatives.

Overall Coordinator - Sunshine (Counselling Cell), IIT Hyderabad

Led Sunshine by coordinating mentors, planning mental wellness initiatives, and organizing institute-wide outreach events.

PR, Outreach & Marketing Coordinator - Finance and Consulting Club (FCC)

Planned and executed publicity, outreach, and marketing campaigns to maximize participation in FCC events.

Hospitality Coordinator - Inter IIT Sports Meet

Coordinated hospitality arrangements for athletes, officials, and guests during the Inter IIT Sports Meet.

RELEVANT COURSES

Computer Science: Data Structures & Algorithms, Theory of Computation, Database Management Systems, Artificial Intelligence, Operating Systems, Introduction to Programming, Discrete Mathematics for Computer Science, Introduction to Computing.

Mathematics: Calculus-I, Calculus-II, Elementary Linear Algebra, Differential Equations, Transform Techniques, Introduction to Probability, Introduction to Statistics, Complex Variables, Introduction to Number Theory.

Misc: Numerical Methods, Digital Fabrication, Probability & Random Variables, Engineering Drawing.

SKILLS

Programming Languages: C, C++, Python, SQL, JavaScript

Web Technologies: HTML, CSS, React, Vite, Flask, REST APIs

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS)

Tools & Libraries: Git, GitHub, Linux, NumPy, Pandas, gmpy2, Solid Edge

PROJECTS

MiniSTL

Tech Stack: C++ OOP Templates RAII Iterator Design HashMap

- Architected a header-only, template-based C++ library from first principles, implementing type-safe **Vector**, **Stack**, **Queue**, **Linked List**, and **HashMap** using abstraction, encapsulation, and operator overloading.
- Engineered mutable and const iterator abstraction layers, decoupling algorithms from container internals, and implemented exponential capacity growth with reserve()/shrink_to_fit() for amortized **O(1)** insertion.
- Designed a **HashMap** with **O(1)** average-case lookup, strictly enforced the **Rule of Five** with **RAII** for exception safety, and built a **Makefile**-driven test suite for one-command verification.

Delivery Logistics Management System

Tech Stack: C++, STL, OOP, File I/O, Git

- **Designed** a modular logistics management system using **object-oriented principles** and real-world entity modeling.
- **Built** a persistence layer using **file I/O** to restore inventory, customers, and orders across executions.
- **Implemented** an end-to-end **order fulfillment workflow** covering inventory validation, approval, truck assignment, delivery, and status tracking.
- **Utilized** STL containers such as **unordered_map** and **vector** for efficient catalog management and scalable object relationships.

Producer-Consumer Synchronization System (Course Project)

Tech Stack: C, Linux, System V Shared Memory, System V Semaphores, IPC

- **Developed** a multi-process **Producer-Consumer** system in C using **System V IPC** mechanisms.
- **Implemented** a shared circular buffer with **shared memory** for efficient inter-process communication.
- **Synchronized** concurrent producer and consumer processes using **System V semaphores** to ensure mutual exclusion and prevent race conditions.
- **Managed** process creation, synchronization, and resource cleanup using **fork()**, **wait()**, and Linux system calls.

ACHIEVEMENTS

Solved **300+** problems in **LeetCode**